arXiv:2304.03373v1 [cs.CV] 6 Apr 2023

Training-Free Layout Control with Cross-Attention Guidance

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silent-chen.github.io/layout-guidance

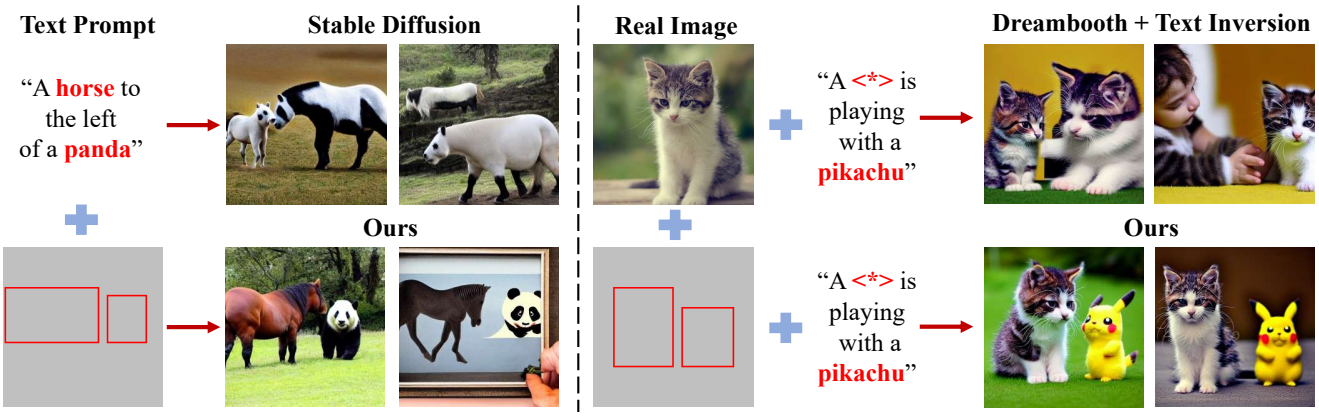


Figure 1: Left: our method can control the layout of an image generated by a pre-trained diffusion model, such as Stable Diffusion [31], without any finetuning or training. It also alleviates the problem that Stable Diffusion may not generate all objects in the prompt. Right: given a single real image, we manage to edit its position, context with the Dreambooth [33], and Text Inversion [8] technique. $\langle * \rangle$ is the learned token that encodes the object in the image.

Abstract

Recent diffusion-based generators can produce high-quality images based only on textual prompts. However, they do not correctly interpret instructions that specify the spatial layout of the composition. We propose a simple approach that can achieve robust layout control without requiring training or fine-tuning the image generator. Our technique, which we call layout guidance, manipulates the cross-attention layers that the model uses to interface textual and visual information and steers the reconstruction in the desired direction given, e.g., a user-specified layout. In order to determine how to best guide attention, we study the role of different attention maps when generating images and experiment with two alternative strategies, forward and backward guidance. We evaluate our method quantitatively and qualitatively with several experiments, validating its effectiveness. We further demonstrate its versatility by extending layout guidance to the task of editing the layout and context of a given real image.

1. Introduction

Generative AI is easily one of the most disruptive technologies that have emerged in the last few years. In computer vision, new text-conditioned image generation methods, such as DALL-E [29], Imagen [34], and Stable Diffusion [31], have demonstrated that machines can generate image outputs of a quality sufficient to use in many applications. These methods can easily multiply the productivity of professional artists as well as laypeople.

Despite these successes, however, many practical applications of image generation, particularly in a professional context, require a high level of *control* that current methods lack. Specifications in language-based image generators are textual; and while text can tap into a vast library of high-level concepts, it is a poor vehicle for expressing fine-grained visual details in an image. In particular, text is a poor way of describing the precise *layout of a composition*.

In fact, as shown in previous works [9] and as we show in our experiments, state-of-the-art image generators are not even very good at correctly interpreting simple layout in-